7 INSULATION

7.1 Scope

The scope of this section comprises the supply and application of insulation conforming to these specifications. The insulation material shall be Closed Cell Elastomeric Nitrile Rubber / Polyethylene Foam / EPDM

7.2 Material

Comply

Thermal insulation material for Duct & Pipe insulation shall be with factory laminated black fiber glass cloth closed cell Elastomeric UV resistant. Thermal conductivity as per BS 874 part 2 – 86 (DIN 52613 52612) / DIN EN 12667 / EN ISO8497 of the insulation material shall not exceed 0.038 W/m°K or 0.212 BTU / (Hr-ft^2-°F/inch) at an average temperature of 30°C. Density of the nitrile rubber shall be 40-60 Kg/m^3, for EPDM shall be 40-60 Kg/m^3 & for polyethylene material it shall be 25-30 Kg/m^3. The product shall have temperature range of –40 °C to 105°C. The insulation material shall be fire rated for Class 0 as per BS 476 Part 6 : 1989 for fire propagation test and for Class 1 as per BS 476 Part 7, 1987 for surface spread of flame test. Water vapour permeability shall be not less than 0.61 per mm (2.48 x 10^{-13} Kg/m.s.Pa i.e. $\mu \geq 7000$: Water vapour diffusion resistance) as per DIN 53122 part 2, DIN 52615 / EN 12086 & EN13469.

Comply

In addition to above properties the insulation material for ducts shall be anti-microbial. Microbiological growth on insulation surface shall be in accordance with ASTM G-21 and bacterial resistance to ASTM2180.

The Material shall comply to ISO 5659 / BS 6853 / ABD 0031 for smoke density and toxicity values. The thermal conductivity of insulation material shall not be affected by aging as per DIN 52616 standard.

Insulation shall be with self-adhesive for ducting and piping and available in roles / sheets.

Comply

Thickness of the insulation shall be as specified for the individual application. Each lot of insulation material delivered at site shall be accompanied with manufacturer's test certificate for density and thickness. Samples of insulation material from each lot delivered at site may be selected by Owner's site representative and gotten Estested for thermal conductivity and density at Contractor's cost. Adhesive used for sealing the insulation shall be non-flammable and with low VOC content (less than 850 gm/l as per USGBC (LEED) and IGBC guidelines) strictly as per manufacturer's recommendations.

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Ducting insulation thickness shall be as per table below.

Noted

Ducting position	Thk. for coastal places
SA duct in RA path	19 mm
Ducted return air system	SA duct: 25 mm RA duct: 19 mm
Both SA & RA exposed	Both 25 mm
SA duct having open cell nitrile rubber insulation	9 mm

Anti-microbial duct insulation material thickness can be 1 mm less than corresponding standard nitrile rubber insulation.

7.3 Duct Insulation

External thermal insulation shall be provided as follows:

The thickness of insulation material shall be as shown on drawings or identified in the schedule of quantity. Following procedure shall be adhered to:

Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work. Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 90^0 cut in thickness of sheet. Hacksaw or blades are not acceptable tools for cutting the insulation.

Noted

Material shall be fitted under compression and no stretching of material shall be permitted. All longitudinal and transverse joints shall be sealed by providing 50 mm wide Fibre glass cloth laminated tape as per manufacturer recommendations. The insulation installers shall be certified by manufacture.

Where ducts/pipes penetrates walls / floor it shall be insulated with intumescent properties insulation material for fire protection. The treatment shall be minimum 500 mm extended on both sides.

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